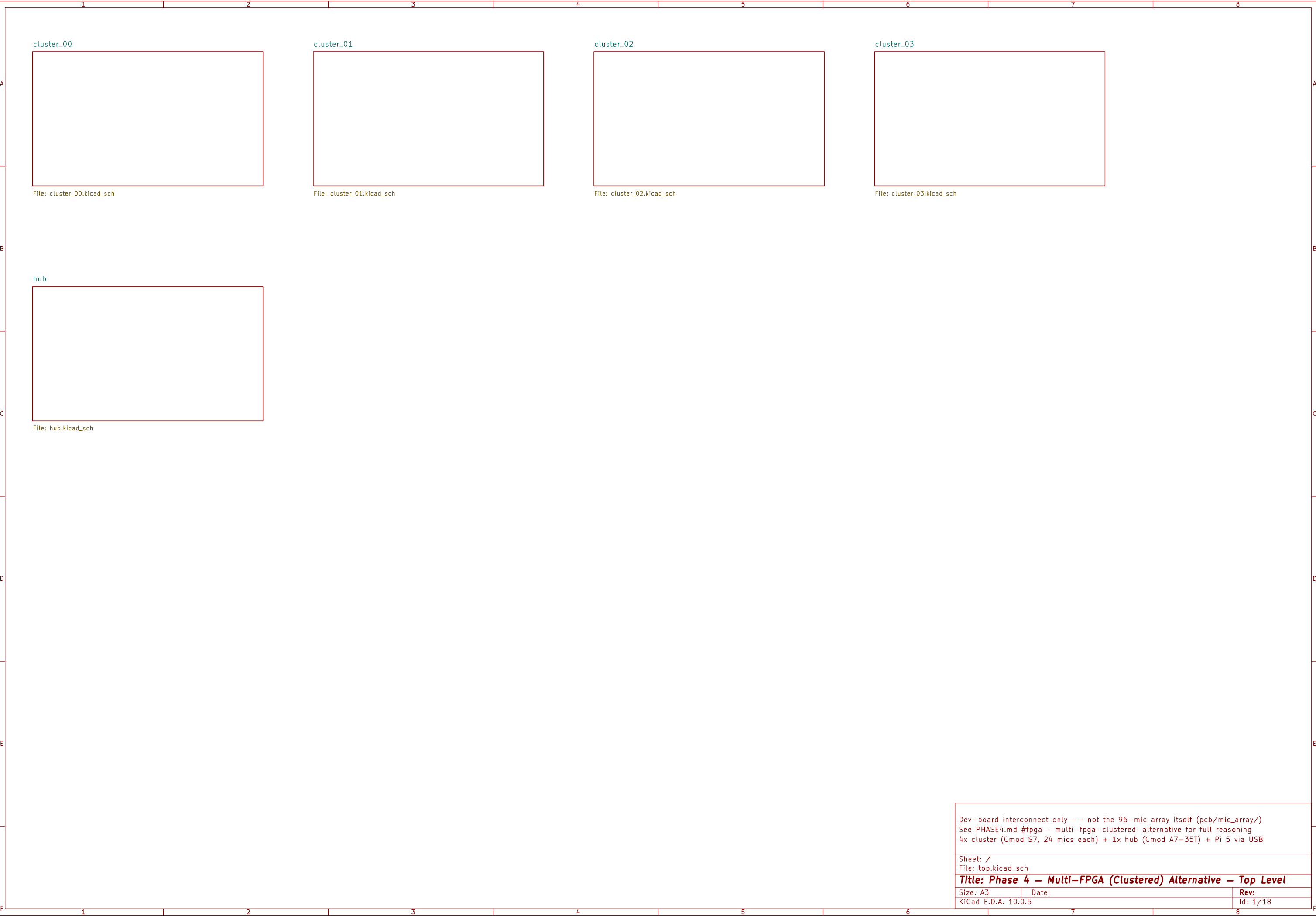
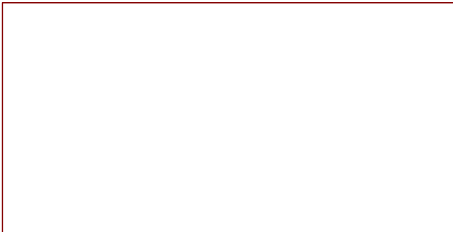




arm_00



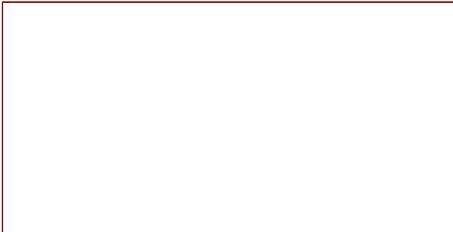
File: arm_00.kicad_sch

arm_01

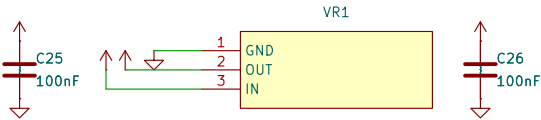
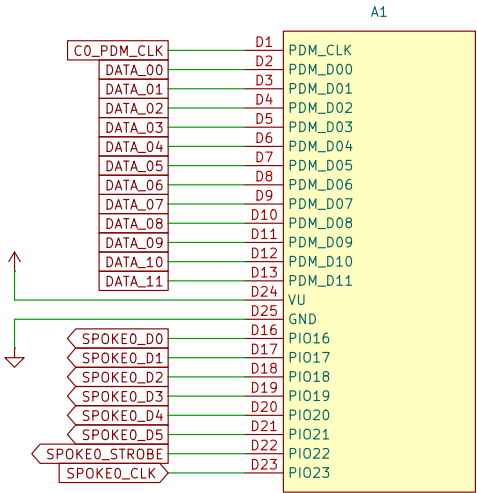


File: arm_01.kicad_sch

arm_02

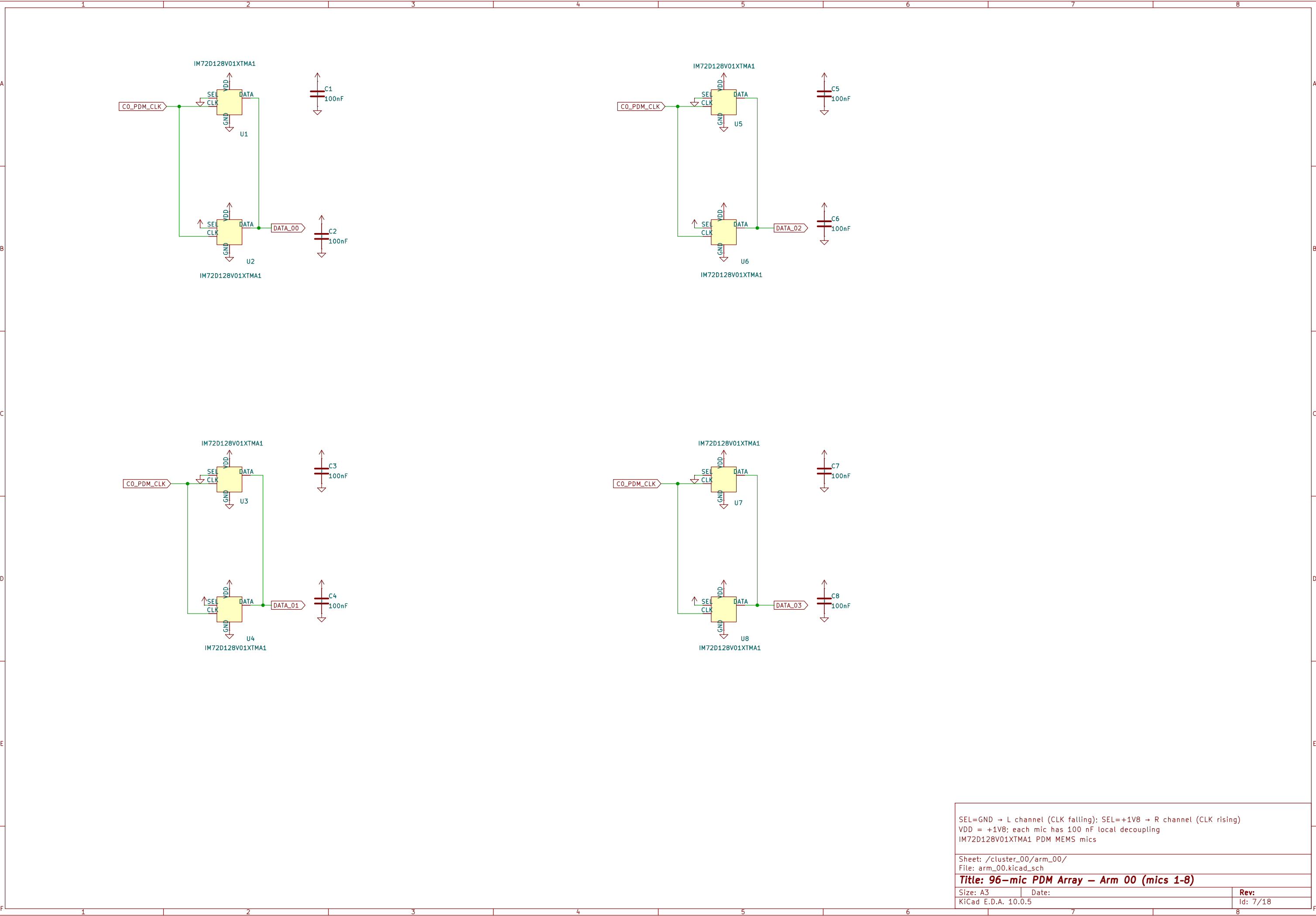


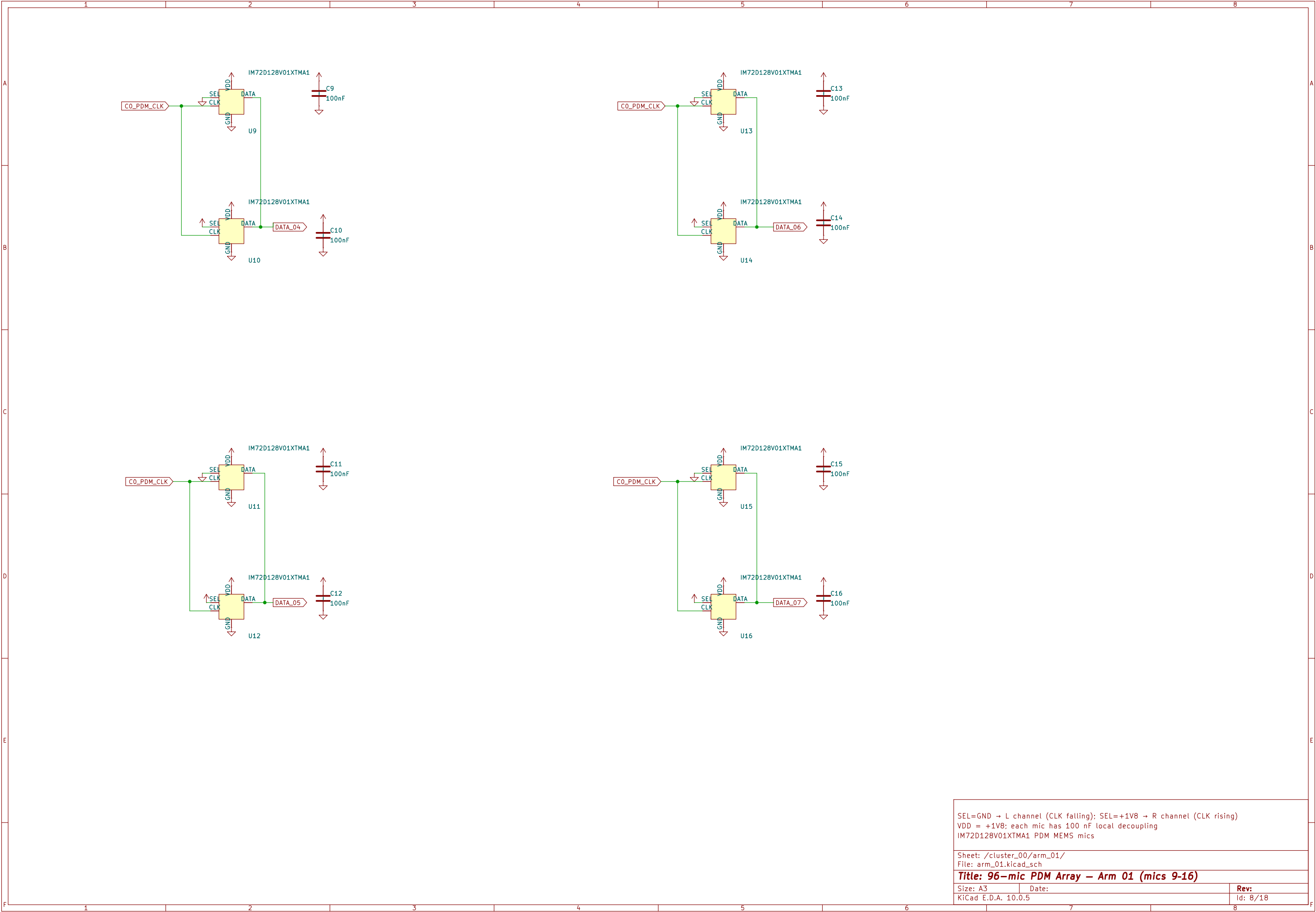
File: arm_02.kicad_sch

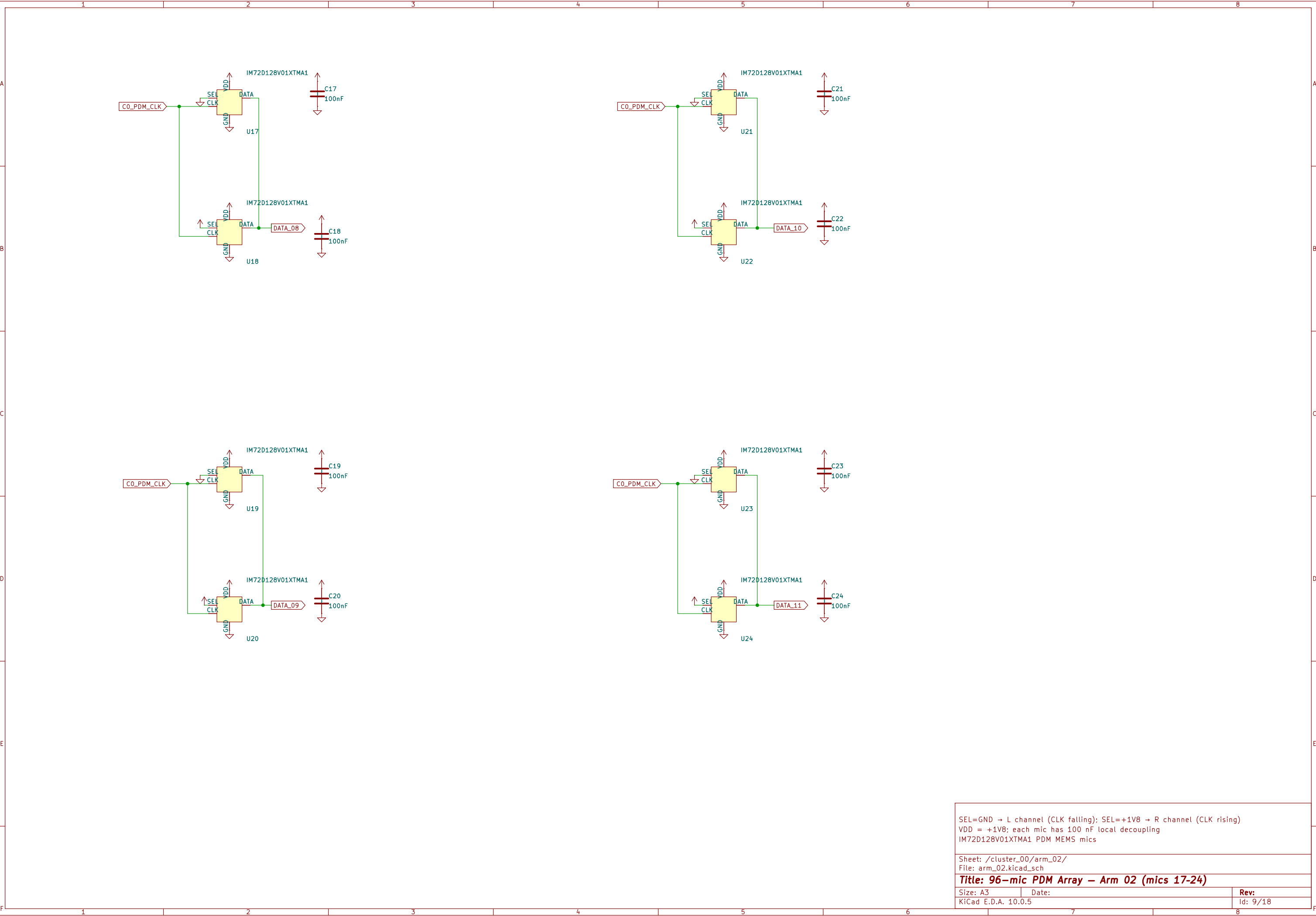


PDM: 12 data lines (2 mics/line) + 1 clock, to this cluster's 3 arms
Spoke bus: 6 data + strobe + fwd clock, parallel single-ended (not LVDS)
Digilent Cmod S7 (XC7S25) – CIC/FIR done in HDL, not shown here

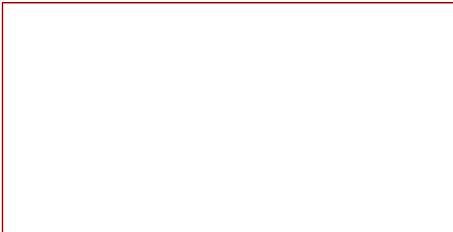
Sheet: /cluster_00/ File: cluster_00.kicad_sch		
Title: Multi-FPGA Alternative – Cluster 0 (arms 0–2, mics 0–23)		
Size: A3	Date:	Rev:
KiCad E.D.A. 10.0.5		Id: 2/18







arm_03



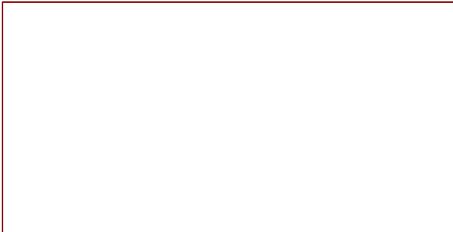
File: arm_03.kicad_sch

arm_04

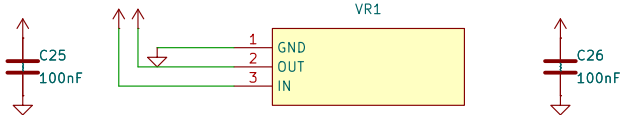
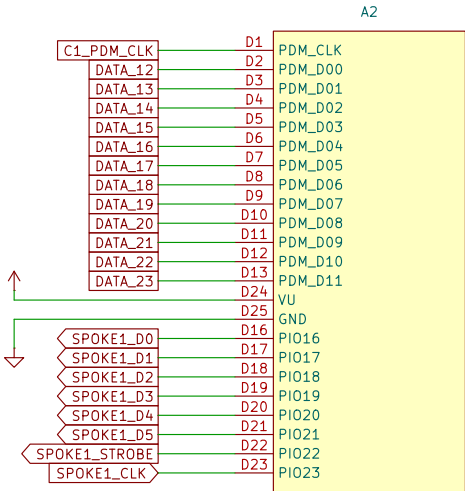


File: arm_04.kicad_sch

arm_05

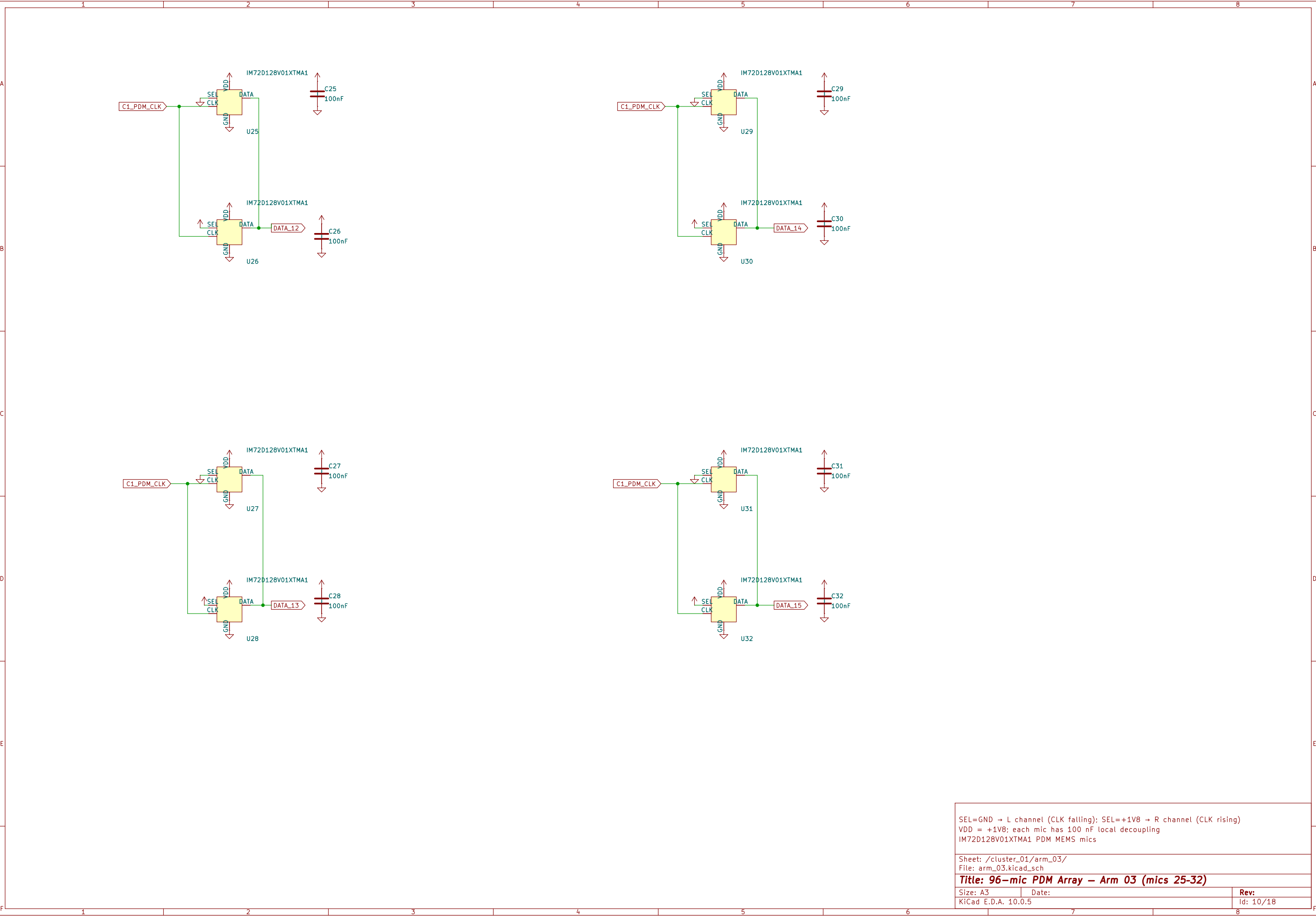


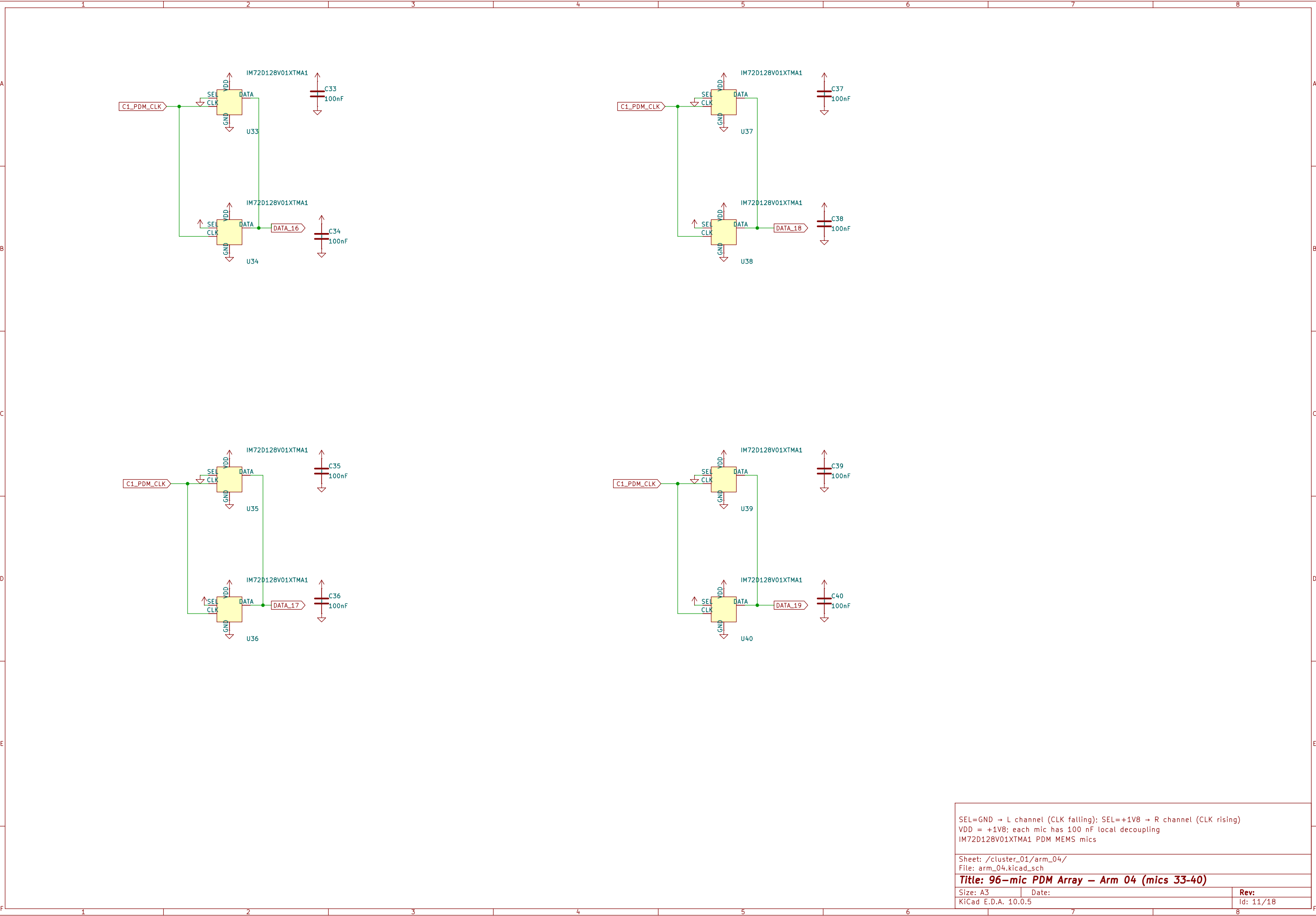
File: arm_05.kicad_sch

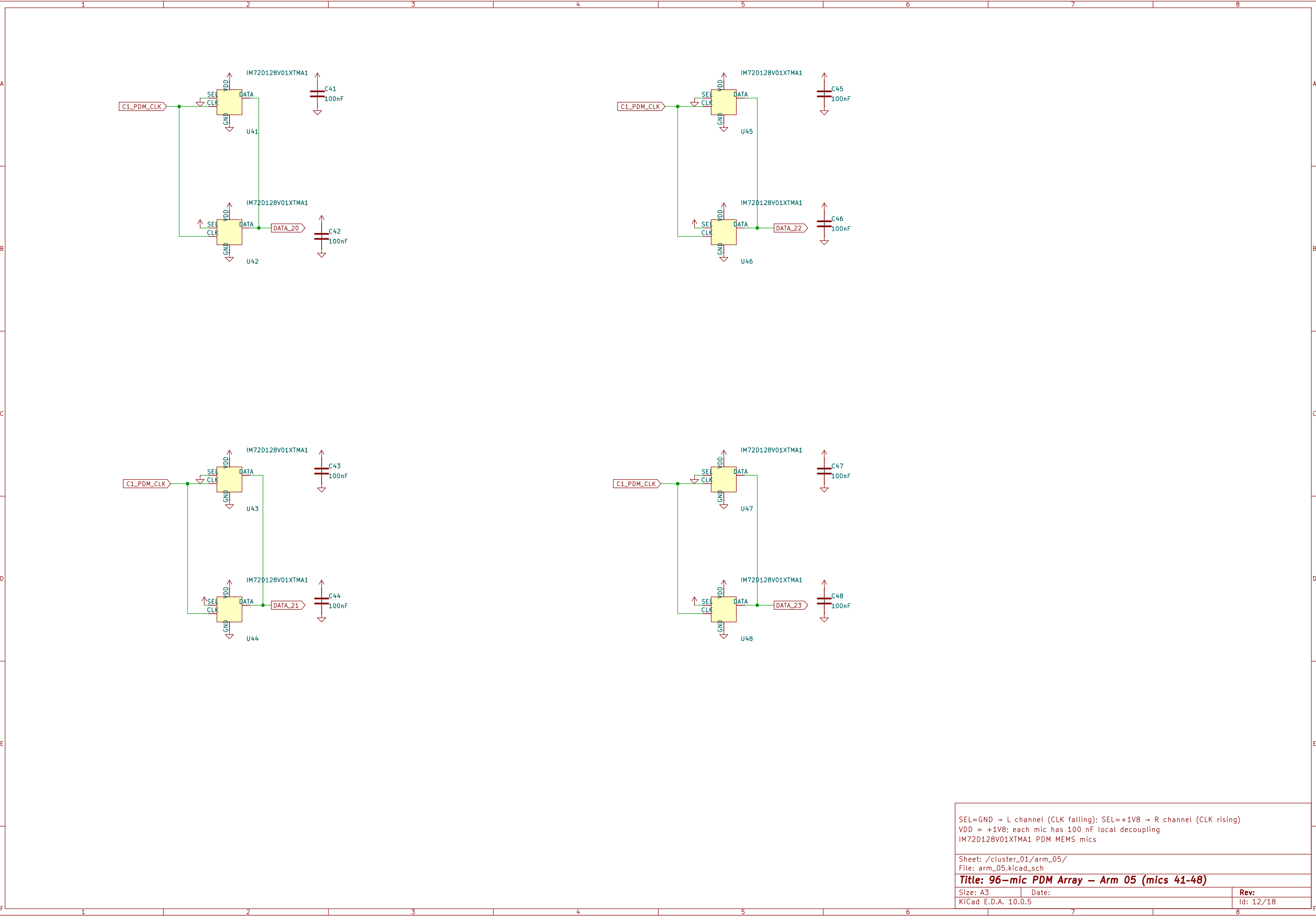


PDM: 12 data lines (2 mics/line) + 1 clock, to this cluster's 3 arms
Spoke bus: 6 data + strobe + fwd clock, parallel single-ended (not LVDS)
Digilent Cmod S7 (XC7S25) – CIC/FIR done in HDL, not shown here

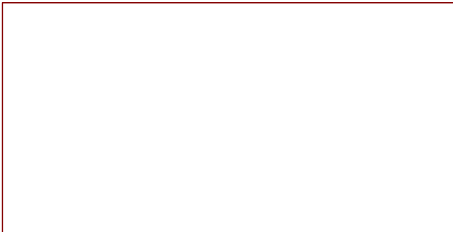
Sheet: /cluster_01/ File: cluster_01.kicad_sch		
Title: Multi-FPGA Alternative – Cluster 1 (arms 3–5, mics 24–47)		
Size: A3	Date:	Rev:
KiCad E.D.A. 10.0.5		Id: 3/18







arm_06



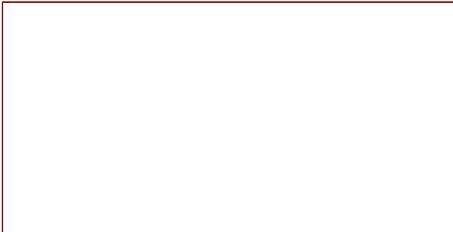
File: arm_06.kicad_sch

arm_07

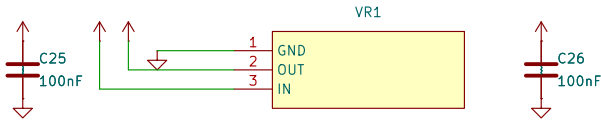
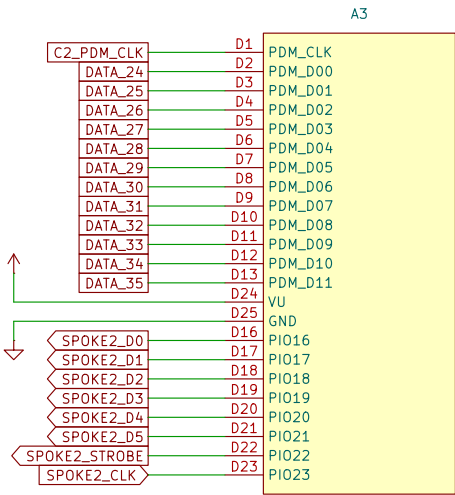


File: arm_07.kicad_sch

arm_08

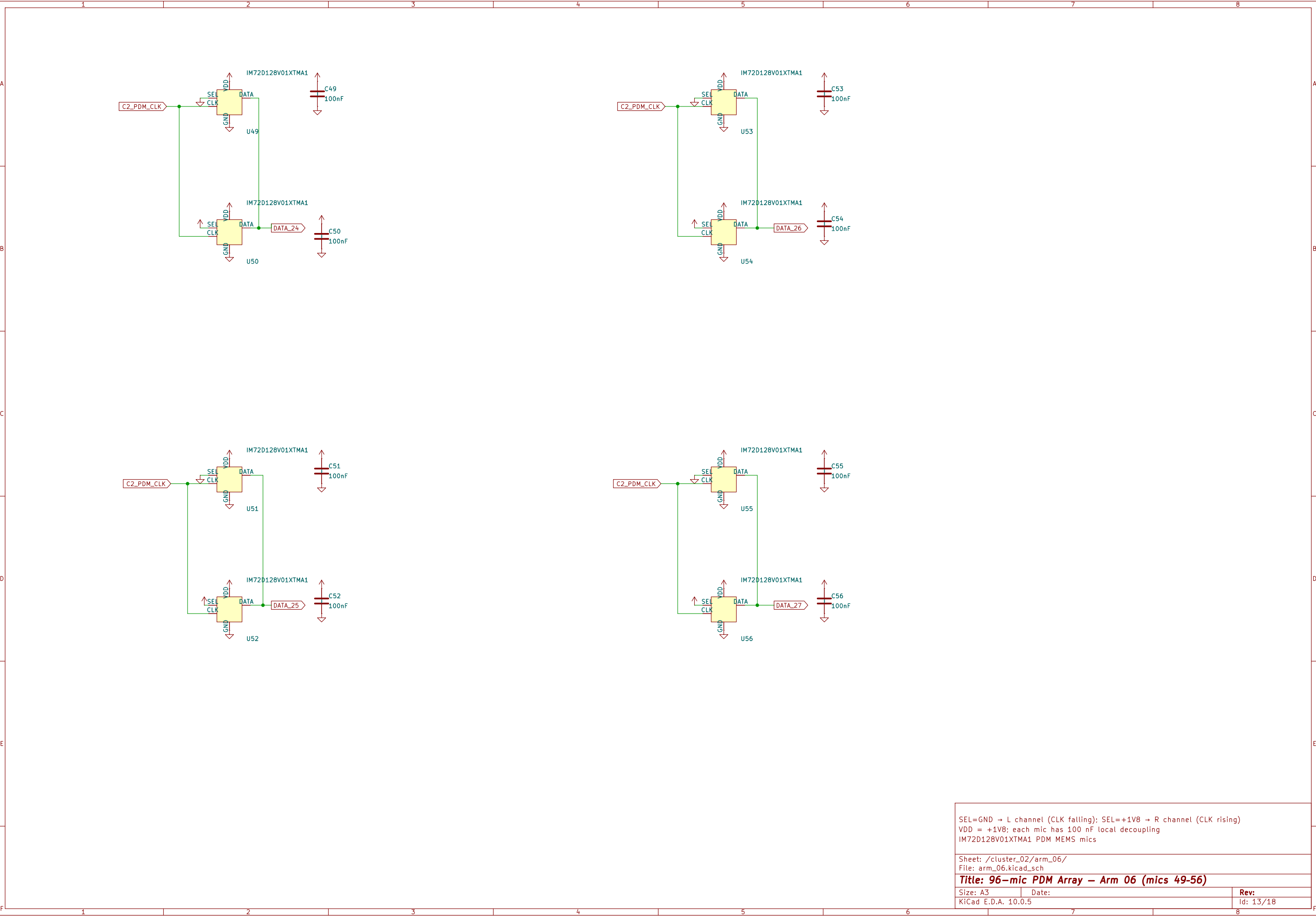


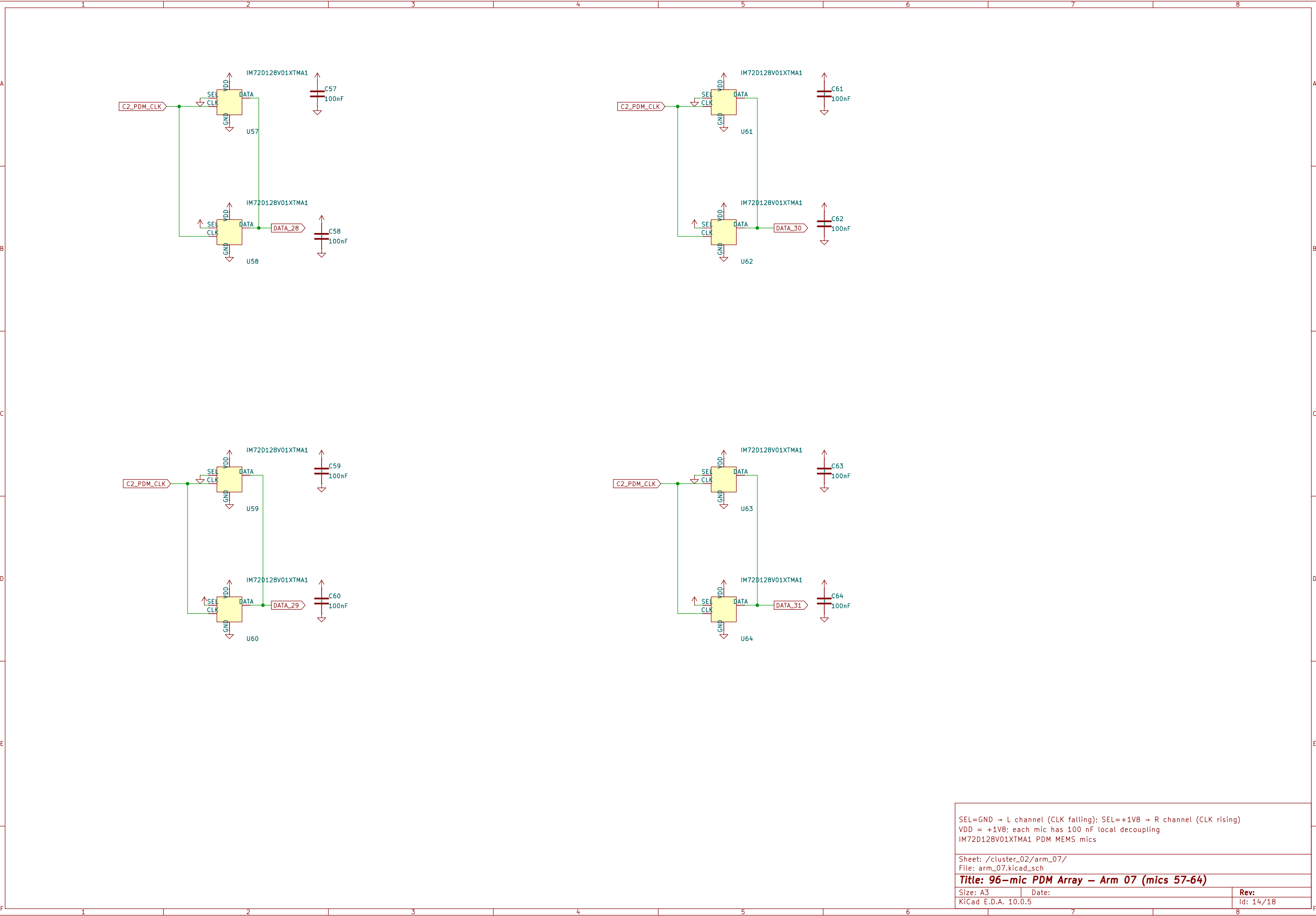
File: arm_08.kicad_sch

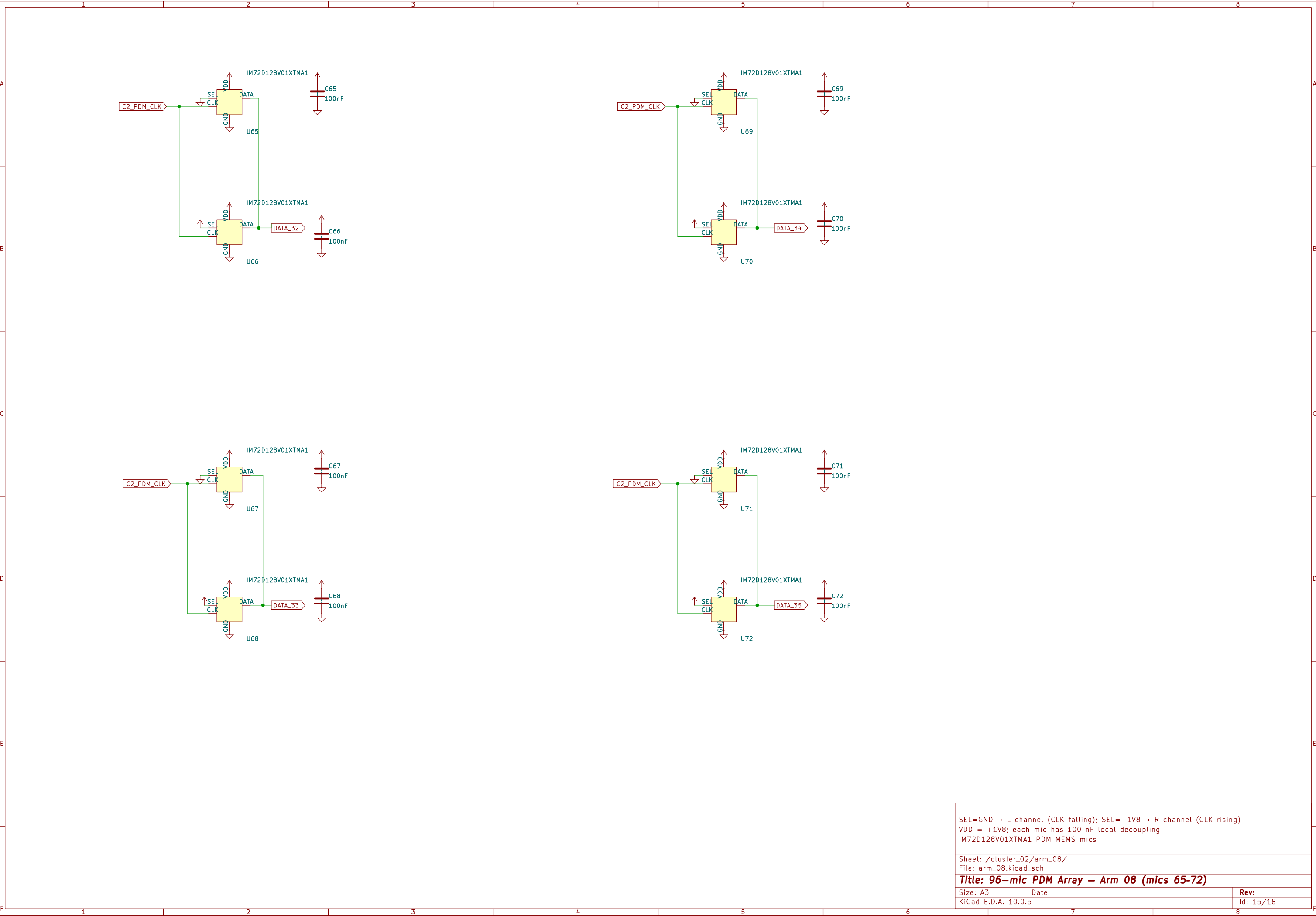


PDM: 12 data lines (2 mics/line) + 1 clock, to this cluster's 3 arms
Spoke bus: 6 data + strobe + fwd clock, parallel single-ended (not LVDS)
Digilent Cmod S7 (XC7S25) – CIC/FIR done in HDL, not shown here

Sheet: /cluster_02/ File: cluster_02.kicad_sch		
Title: Multi-FPGA Alternative – Cluster 2 (arms 6–8, mics 48–71)		
Size: A3	Date:	Rev:
KiCad E.D.A. 10.0.5		Id: 4/18

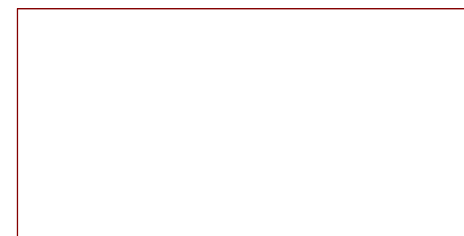




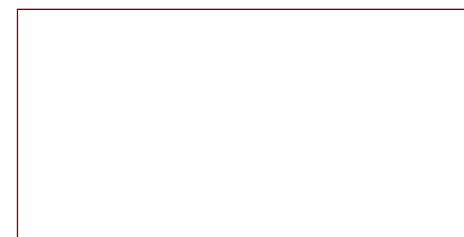


SEL=GND → L channel (CLK falling); SEL=+1V8 → R channel (CLK rising)
VDD = +1V8; each mic has 100 nF local decoupling
IM72D128V01XTMA1 PDM MEMS mics

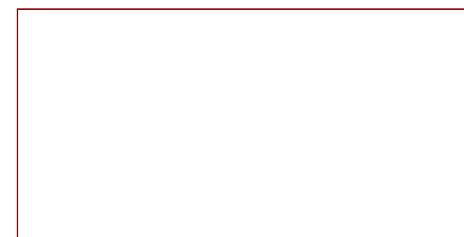
Sheet: /cluster_02/arm_08/ File: arm_08.kicad_sch		
Title: 96-mic PDM Array – Arm 08 (mics 65-72)		
Size: A3	Date:	Rev:
KiCad E.D.A. 10.0.5	Id: 15/18	



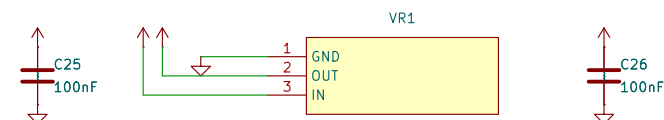
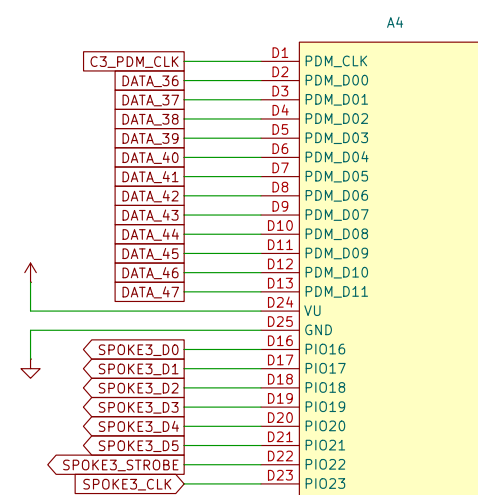
File: arm_09.kicad_sch



File: arm_10.kicad_sch



File: arm_11.kicad_sch

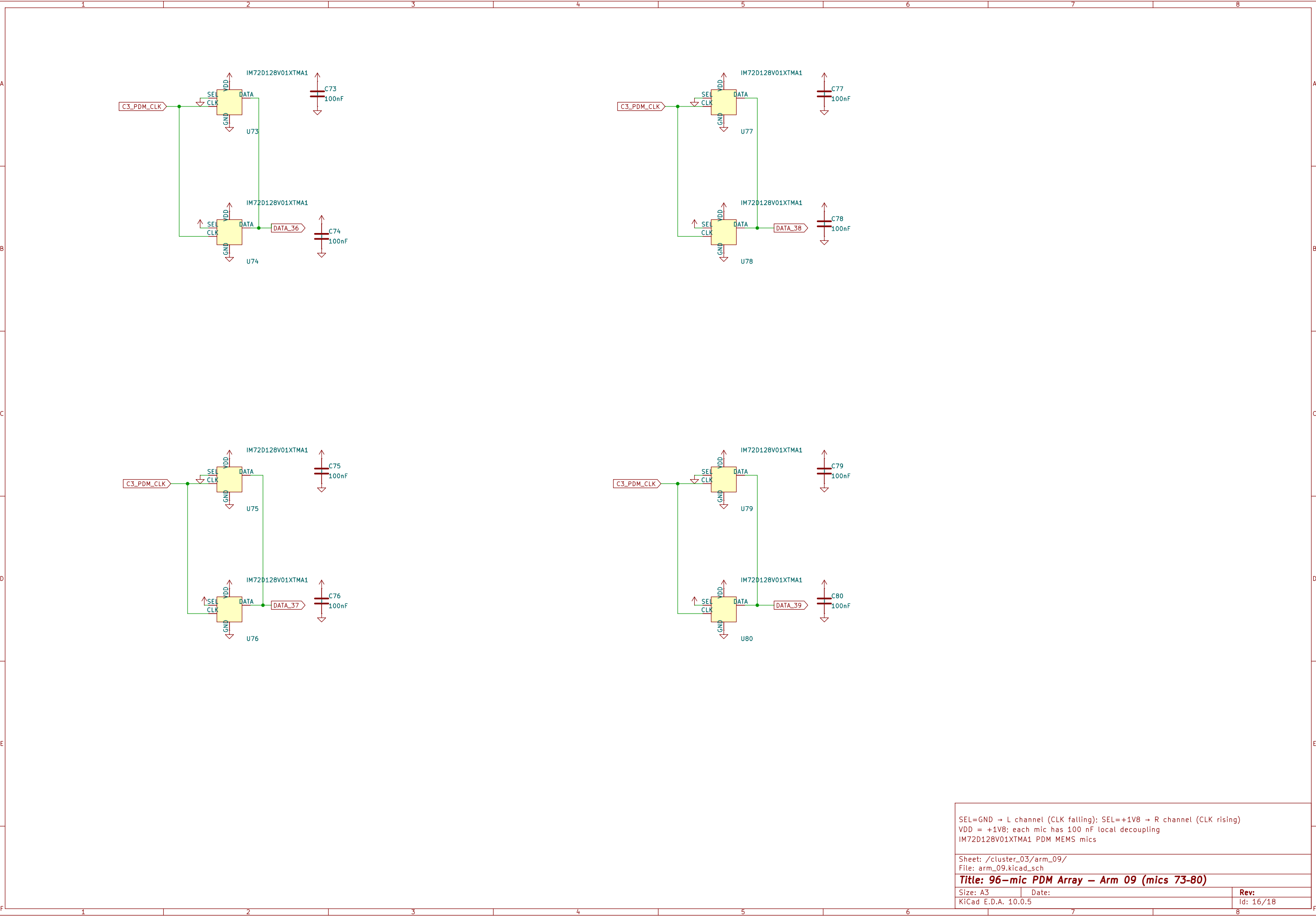


PDM: 12 data lines (2 mics/line) + 1 clock, to this cluster's 3 arms
Spoke bus: 6 data + strobe + fwd clock, parallel single-ended (not LVDS)
Digilent Cmod S7 (XC7S25) – CIC/FIR done in HDL, not shown here

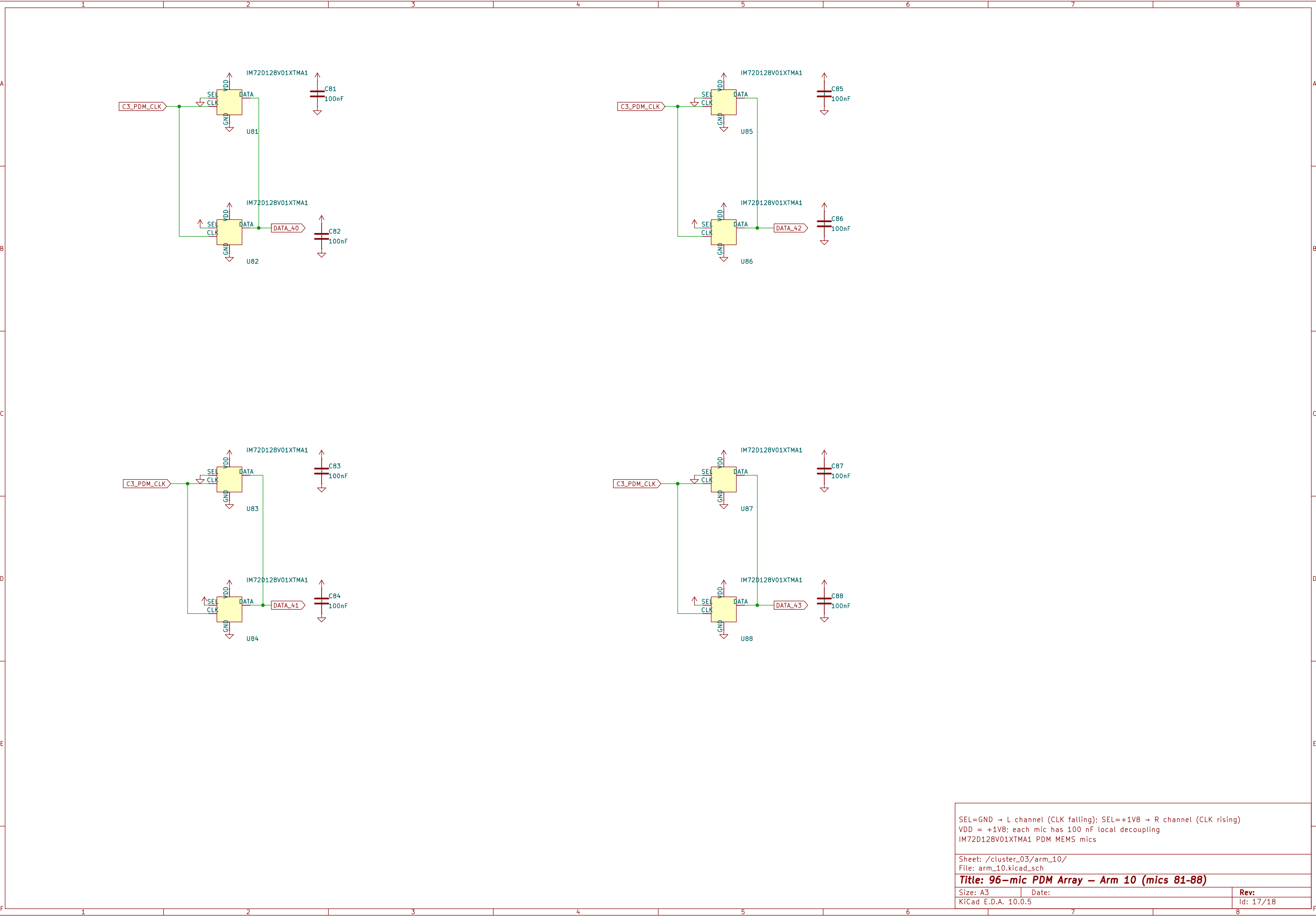
Sheet: /cluster_03/
File: cluster_03.kicad_sch

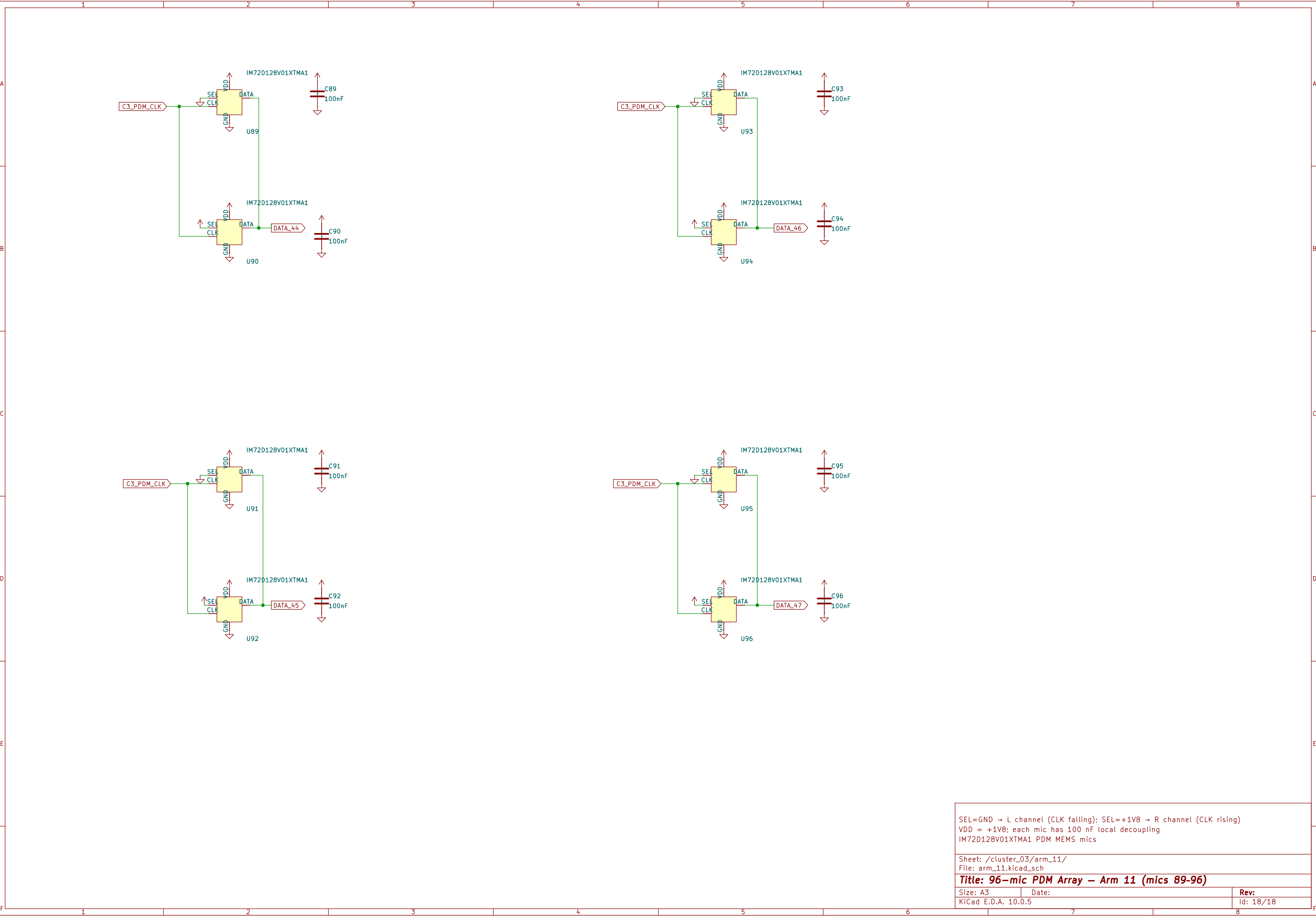
Title: Multi-FPGA Alternative – Cluster 3 (arms 9–11, mics 72–95)

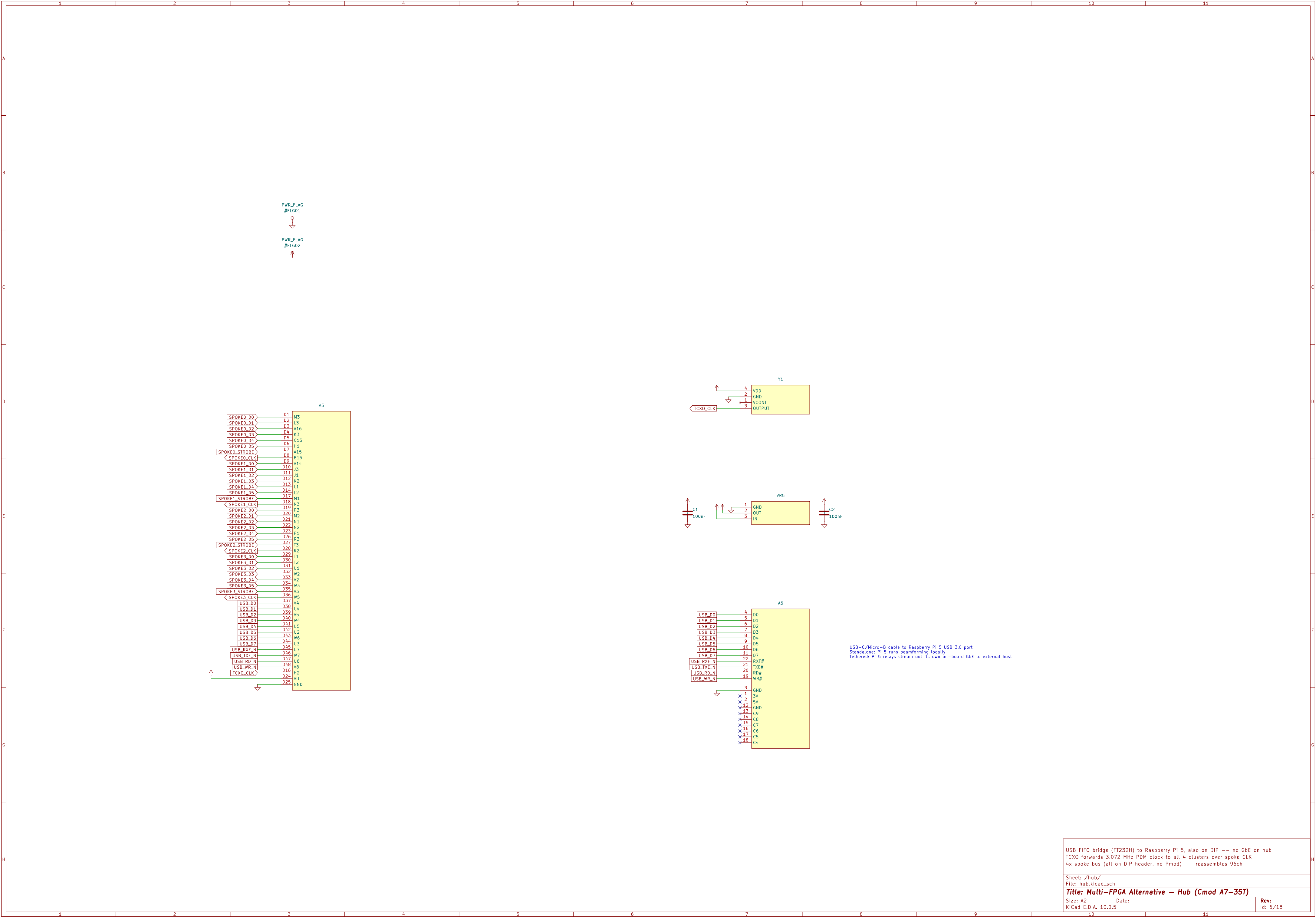
Size: A3	Date:	Rev:
KiCad E.D.A. 10.0.5		Id: 5/18



SEL=GND → L channel (CLK falling); SEL=+1V8 → R channel (CLK rising) VDD = +1V8; each mic has 100 nF local decoupling IM72D128V01XTMA1 PDM MEMS mics		
Sheet: /cluster_03/arm_09/ File: arm_09.kicad_sch		
Title: 96-mic PDM Array – Arm 09 (mics 73-80)		
Size: A3	Date:	Rev:
KiCad E.D.A. 10.0.5		Id: 16/18







USB FIFO bridge (FT232H) to Raspberry Pi 5, also on DIP -- no GbE on hub
TCXO forwards 3.072 MHz PDM clock to all 4 clusters over spoke CLK
4x spoke bus (all on DIP header, no Pmod) -- reassembles 96ch